

Digital Communications - NUIST

Engineering Technology

May, 2026



Short Title:	Digital Communications (NUIST)	
Faculty	Science and Computing	
Department	Engineering Technology	
Cluster	Discipline Specific Technologies	
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Credits	5	Level: Postgraduate

Description of Module / Aims

Digital Communications builds on a theory typically covered in an undergraduate programme in basic modulation, propagation and signal recovery techniques over wireless, copper and optical media. The module deals with error detecting and correcting codes, multi-carrier systems, signal recovery in a digital system and optical communications from fundamentals to Wavelength Division Multiplexing (WDM).

Programmes

COMM-0003	MEng in Electronic Engineering (WD_EELEN_R)	1 / 2 / M
	MEng in Electronic Information Systems (WD_EELEN_RI)	2 / 4 / M
COMM-0003	PG Dip in Engineering in Electronic Engineering (WD_EELEN_G)	1 / 2 / M

Indicative Content

- Signal recovery: digital filtering with adaptive equalisation
- Error detection (cyclic codes) and error correcting codes using linear block codes
- Orthogonal multi-carrier communications over wireless and copper media
- Optical Communications (components, budgetting, Wavelength Division Multiplexing operation and impact of non-linear behaviour, Four wave mixing and Cross Phase Modulation)

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Contrast the different approaches to and successes of, both in theory and practice, error detecting and correcting codes.
2. Analyse the performance defining parameters of orthogonal, multi-carrier communications and how these can be effected under various scenarios.
3. Assess the merits and operation in practice of optical communications, defining the key relevant parameters and their relationships.
4. Examine and critique the theory and practical techniques in adaptive digital filtering.
5. Demonstrate a practical ability to develop, troubleshoot, test, analyse and report on Digital Communications.

Learning and Teaching Methods

- In-class: Lectures, Group Discussion, Practical exercises
- Online: Lecture notes, Laboratory descriptions, Assignments; Discussion forum driven by lecturer
- Workshop exercises

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1, 2, 3, 4
Continuous Assessment	50%	
<i>Project</i>	50%	1,2,3,4,5

Assessment Criteria

70% -

100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

40% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Workshop	6	
Independent Learning	105	

Supplementary Material

- Open University, .. _Digital Communications_ . eBook: The Open University, March 1, 2016.
- Proakis, J. and S. Masoud. _Digital Communications_. London: McGraw Hill, 2007.
- Sklar, B. _Digital Communications_. New Jersey: Prentics Hall, 2009.